ADVANCED NUTRITIONAL BIOCHEMISTRY & PHARMACOGNOSY

Biological Functionality of Cruciferous Vegetables, Fruits, Herbs & Spices

Molecular Receptors, Detoxification Cascades, Bioavailability Kinetics & Clinical Mechanisms

CRUCIFEROUS

Sulforaphane & DIM

Nrf2-Keap1 induction, Phase II enzymes & estrogen metabolite shift.

TROPICAL & SUBTROPICAL

Bromelain & Hesperidin

Proteolytic cleavage, eNOS stimulation & Urolithin A mitophagy.

TEMPERATE FRUITS

Anthocyanins & Quercetin

Blood-brain barrier transit, BDNF & 2.5-yr cognitive delay.

HERBS & SPICES

Curcumin & Capsaicin

Lamiaceae volatiles, +2000% piperine surge & TRPV1 analgesia.

Curriculum: Functional Food Science & Clinical Phytotherapy | Scientific Source: Manach et al. (2004) Am. J. Clin. Nutr. 79(5):727-747

PHYTOCHEMICAL TAXONOMY

Classification of Dietary Phytochemicals & Molecular Targets

Key Biological Action: Four major structural classes orchestrate pleiotropic cell signaling in human physiology.

Glucosinolates & Isothiocyanates

BRASSICACEAE SPECIFIC

- Sulfur-containing glucosides hydrolyzed by plant myrosinase.
- Yield bioactive isothiocyanates: Sulforaphane, AITC, BITC.
- Potent Phase II enzyme inducer via Nrf2-Keap1 electrophilic binding.
- Direct anti-microbial and cytoprotective capabilities.

Polyphenols & Flavonoids

FRUITS, HERBS & TEA

- Flavanones (Hesperidin), Anthocyanins, Flavonols (Quercetin).
- Ellagitannins (Punicalagin) metabolized to Urolithin A.
- Direct radical scavenging, metal ion chelation, and eNOS stimulation.
- Inhibition of pro-inflammatory NF-κB nuclear translocation.

Terpenoids, Esters & Alkaloids

HERBS & PUNGENT SPICES

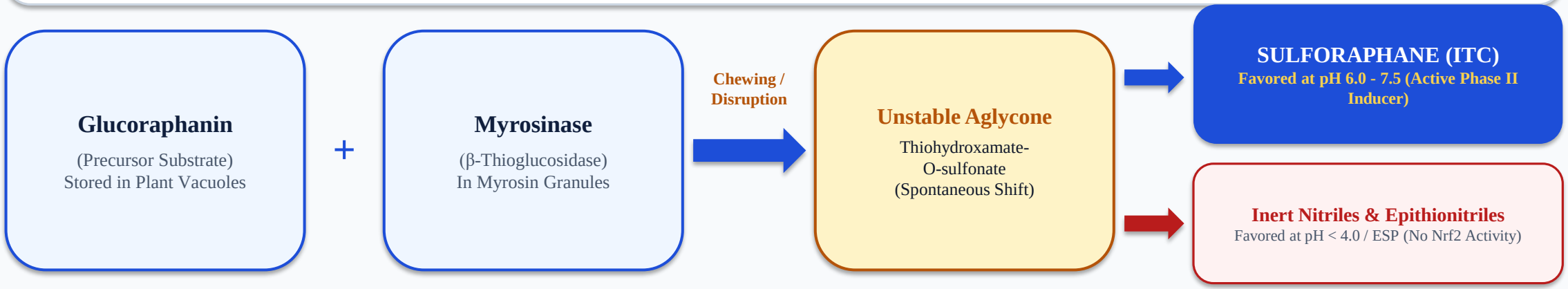
- Monoterpenes (Thymol, Carvacrol) & Abietanes (Carnosic acid).
- Phenolic diarylheptanoids (Curcumin) in Curcuma longa.
- Vanilloids (Capsaicin) activating thermo-sensitive TRPV1 channels.
- Alkaloids (Piperine) non-competitively suppressing hepatic UGTs.

Scientific Reference: Dinkova-Kostova, A. T., & Kostov, R. V. (2012). Glucosinolates and isothiocyanates in health and disease. Trends in Molecular Medicine, 18(6), 337-347. DOI: 10.1016/j.molmed.2012.04.003

The Glucosinolate-Myrosinase Enzyme Cascade

Key Biological Action: Physical cellular disruption is required to convert inert glucosinolates into bioactive isothiocyanates.

- Glucoraphanin is compartmentalized in vacuoles; myrosinase enzyme resides in separate myrosin cells.
- Mastication ruptures cell walls, triggering rapid hydrolysis into an unstable aglycone intermediate.



Critical Culinary Kinetic Rule: Severe boiling (>60°C) completely denatures heat-labile myrosinase, reducing sulforaphane yield by 85%. In contrast, gentle steaming for 3 to 4 minutes preserves enzyme integrity while softening cell walls, maximizing bioactive delivery upon ingestion.

Scientific Reference: Fahey, J. W., & Talalay, P. (1997). Antioxidant functions of sulforaphane: a potent inducer of Phase II detoxication enzymes. Food and Chemical Toxicology, 37(9-10), 973-979. DOI: 10.1016/S0278-6915(99)00082-4

Sulforaphane: Nrf2-Keap1 Induction & Phase II Detoxification

Key Biological Action: Sulforaphane is the most potent naturally occurring inducer of the cytoprotective Nrf2-ARE gene pathway.

STEP 01

Basal Repression

- Keap1 homodimer sequesters Nrf2 in the cytoplasm.
- Keap1 acts as substrate adaptor for Cul3-Rbx1 ubiquitin ligase.
- Nrf2 is constantly ubiquitinated and degraded by 26S proteasome.
- Basal Nrf2 half-life: ~20 min.

STEP 02

Cys151 Sensing

- Sulforaphane's -N=C=S group acts as an electrophile.
- Covalently modifies reactive Cys151 thiol sensors on Keap1.
- Keap1 undergoes conformational structural shift.
- Halts Cul3 ubiquitination; newly translated Nrf2 escapes.

STEP 03

Nuclear Transit

- Free Nrf2 accumulates and translocates across nuclear pores.
- Heterodimerizes with small Maf transcription factors (sMaf).
- Recognizes specific 5'-TGACNNNGC-3' consensus core.
- Recruits transcriptional coactivator complexes (CBP/p300).

STEP 04

ARE Gene Induction

- Binds Antioxidant Response Elements (ARE) in DNA promoters.
- Induces NQO1 (quinone reductase) & GSTs (glutathione transferases).
- Upregulates HO-1 and γ-GCS (glutathione synthesis rate-limiting).
- Enhances cellular defense against xenobiotics and ROS.

Downstream Physiological Outcome: Upregulation of >200 phase II genes elevates intracellular glutathione (GSH) synthesis by >200%, accelerates urinary excretion of airborne pollutants (benzene, acrolein), and halts chemical carcinogen-DNA adduct formation.

Scientific Reference: Kensler, T. W., et al. (2013). Keap1-nrf2 signaling: a target for cancer prevention by sulforaphane. Carcinogenesis, 34(2), 243-252. DOI: 10.1093/carcin/bgs340

Indole-3-Carbinol (I3C) & DIM: Estrogen Metabolism & Epigenetics

Key Biological Action: Acid-catalyzed condensation of I3C produces 3,3'-diindolylmethane (DIM), which modulates steroid metabolism.

Steroid Modulation & Epigenetic Checkpoints

• Gastric Condensation: In stomach acid, I3C polymerizes into 3,3'-diindolylmethane (DIM) and indolo[3,2-b]carbazole.

• Hydroxylation Shifting: DIM selectively induces CYP1A1/CYP1A2 via the Aryl Hydrocarbon Receptor (AhR).

• Hormonal Ratio: Increases protective 2-hydroxyestrone (2-OHE1) while suppressing genotoxic 16α-hydroxyestrone (16α-OHE1).

• Histone Deacetylase (HDAC) Block: Inhibits Class I/II HDACs, restoring p21/p27 expression and arresting cells in G1 phase.

• Dietary Matrix: High abundance in broccoli, Brussels sprouts, kale, and purple cabbage.



Figure 1: High-Glucosinolate Cruciferous Matrices (Brassica oleracea)

Scientific Reference: Bradlow, H. L., et al. (1999). 2-hydroxyestrone: the 'good' estrogen. Journal of Cellular Biochemistry Supplement, 32-33, 144-152. DOI: 10.1002/(SICI)1097-4644(1999)75:32+<144::AID-JCB19>3.0.CO;2-E

Tropical Fruits: Proteolytic Enzymes & Unique Xanthones

Key Biological Action: Papaya, pineapple, mango, and guava deliver specialized biocatalysts and lipophilic carotenoids.

Biocatalytic & Phytochemical Attributes

• Pineapple (Ananas comosus): Concentrates Bromelain, a sulfhydryl proteolytic complex stable across pH 3.0 to 8.5.

• Papaya (Carica papaya): Delivers Papain (cysteine endopeptidase) and free non-esterified β-cryptoxanthin.

• Mango (Mangifera indica): Rich in Mangiferin, a rare C-glucosylxanthone resistant to gastric acid breakdown.

• Guava (Psidium guajava): Superior source of natural ascorbic acid (>200 mg/100g) and all-trans lycopene.

• Therapeutic Utility: Facilitates gastrointestinal protein digestion and reduces systemic mucosal edema.

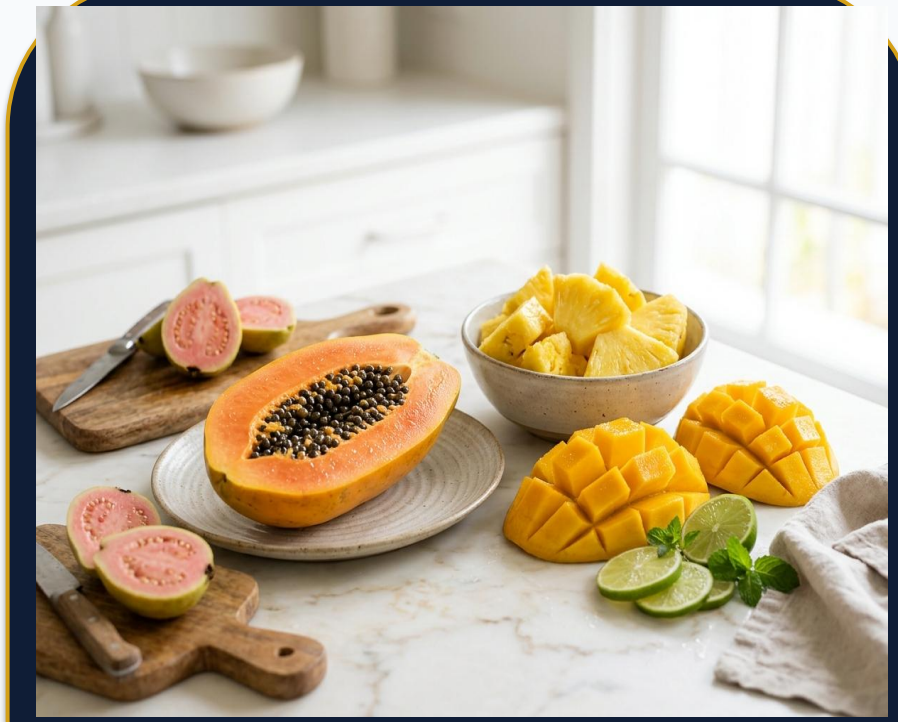


Figure 2: Proteolytic & Xanthone-Rich Tropical Matrices

Scientific Reference: Pavan, R., Jain, S., Shraddha, & Kumar, A. (2012). Properties and therapeutic application of bromelain: a review. Biotechnology Research International, 2012, 976203. DOI: 10.1155/2012/976203

METABOLIC FUNCTIONALITY

Tropical Fruits: Anti-Inflammatory & Metabolic Actions

Key Biological Action: Bromelain and mangiferin target cell adhesion molecules, COX-2, and AMPK energy metabolism.

Bromelain Systemic Anti-Inflammation

• Selectively cleaves CD44 cell surface adhesion molecules, blocking leukocyte migration into inflamed joints.

• Decreases Prostaglandin E2 (PGE2) and Thromboxane A2 via selective inhibition of Cyclooxygenase-2 (COX-2).

• Promotes fibrinolysis by converting plasminogen to active plasmin, reducing tissue edema and vascular exudates.

• Clinical trials demonstrate pain reduction in osteoarthritis comparable to low-dose NSAIDs without gastric lesions.

Mangiferin Glycemic & Lipid Modulation

• Phosphorylates AMP-activated protein kinase (AMPK) in skeletal muscle and hepatocytes.

• Upregulates GLUT4 glucose transporter translocation independently of insulin signaling pathways.

• Competitive inhibitor of intestinal α-glucosidase and pancreatic α-amylase, suppressing postprandial glucose spikes.

• Downregulates sterol regulatory element-binding protein-1c (SREBP-1c), attenuating hepatic steatosis.

Scientific Reference: Imran, M., et al. (2017). Mangiferin: a natural miracle bioactive compound against lifestyle related disorders. Lipids in Health and Disease, 16(1), 84. DOI: 10.1186/s12944-017-0449-y

Subtropical Fruits: Citrus Flavanones & Pomegranate Ellagitannins

Key Biological Action: High concentrations of hesperidin, naringin, and punicalagins dictate cardiovascular protection.

Citrus & Pomegranate Phytochemical Profile

• Citrus Flavanone Glycosides: Hesperidin and naringin concentrated within the albedo (inner peel) and carpel membranes.

• Rhamnosidase Requirement: Colonic bacteria cleave disaccharide sugars, liberating lipophilic aglycones (hesperetin, naringenin).

• Punicalagins (Pomegranate): Giant hydrolyzable ellagitannins (MW ~1084 g/mol) unique to Punica granatum pericarp and arils.

• Lactone Cleavage: Hydrolyze into ellagic acid, serving as the biological substrate for microbial urolithin synthesis.

• Antioxidant Capacity: Total antioxidant capacity (ORAC) of pomegranate exceeds green tea and red wine by threefold.



Figure 3: Flavanone & Ellagitannin-Rich Subtropical Matrices

Scientific Reference: Gattuso, G., Barreca, D., Gargiulli, C., Leuzzi, U., & Caristi, C. (2007). Flavonoid composition of citrus juices. Journal of Agricultural and Food Chemistry, 55(25), 9989-1002. DOI: 10.1021/jf071700a

CARDIOVASCULAR BIOLOGY

Endothelial Protection & Urolithin A-Induced Mitophagy

Key Biological Action: Citrus and pomegranate bioactives enhance arterial compliance and purge damaged mitochondria.

STAGE 01

Ellagitannin Ingestion

- Punicalagins ingested from pomegranate (MW ~1084).
- Spontaneous hydrolysis to Ellagic Acid in gut lumen.
- Limited direct native uptake due to bulky hydrophilicity.

STAGE 02

Microbial Cleavage

- Colonic Gordonibacter urolithinfaciens metabolizes ellagic acid.
- Cleaves lactone rings yielding bioactive Urolithin A.
- Lipophilic metabolite readily enters systemic bloodstream.

STAGE 03

Mitochondrial Target

- Urolithin A homes to depolarized, aging mitochondria.
- Prevents degradation of PINK1 on outer membrane.
- Recruits cytosolic Parkin E3 ubiquitin ligase to surface.

STAGE 04

Mitophagy Clearance

- Ubiquitinated organelles engulfed by autophagosomes.
- Lysosomes degrade defective, ROS-generating organelles.
- Result: Stimulates biogenesis & raises muscle ATP (+12%).

Cardiovascular & Mitophagy Consensus: Citrus hesperidin stimulates eNOS Ser1177 phosphorylation, reducing diastolic blood pressure by 4.5 mmHg in randomized trials. Concurrently, Urolithin A selectively cleanses aging organelles, increasing human muscle endurance without organelle toxicity.

Scientific Reference: Espín, J. C., Larrosa, M., García-Conesa, M. T., & Tomás-Barberán, F. A. (2013). Biological significance of urolithins. Molecular Nutrition & Food Research, 57(7), 1199-1213. DOI: 10.1002/mnfr.201300058

Temperate Fruits: Anthocyanins, Proanthocyanidins & Quercetin

Key Biological Action: Berries, apples, cherries, and grapes provide structural flavylium ions with high radical-quenching capacity.

Flavylium Chromophores & Polyphenolic Tannins

• Anthocyanins (Vaccinium & Rubus): Cyanidin, malvidin, and delphinidin 3-glucosides characterized by the positive flavylium ion.

• Extreme Scavenging: Direct electron donation and hydrogen atom transfer (HAT) neutralizing superoxide and peroxyl radicals.

• A-Type Proanthocyanidins (Cranberry): Possess dual interflavan bonds (C4-C8 & C2-O-C7), preventing uropathogenic P-fimbriae bacterial adhesion.

• Apple Quercetin Glycosides: Highly concentrated in the peel (5x higher than pulp), inhibiting xanthine oxidase and lipid peroxidation.

• Vascular Tonus: Protects microvascular capillary walls against oxidative hyperpermeability.

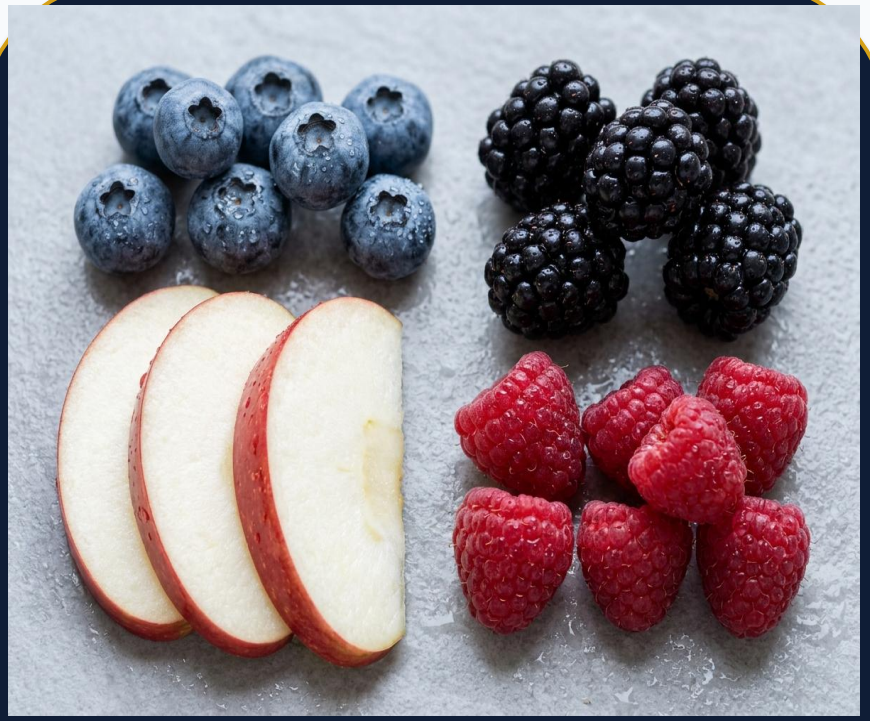


Figure 4: Flavylium-Rich Berries and Temperate Fruits

Scientific Reference: Cassidy, A., O'Reilly, E. J., et al. (2011). Habitual intake of flavonoid subclasses and risk of Parkinson's disease and cardiovascular disease. American Journal of Clinical Nutrition, 93(2), 338-347. DOI: 10.3945/ajcn.110.008102

Temperate Fruits: Neuroprotection & Intestinal Glucose Blunting

Key Biological Action: Anthocyanin metabolites cross the blood-brain barrier and downregulate intestinal glucose absorption.

Blood-Brain Barrier Crossing & Synaptic Plasticity

• Circulating glucuronidated and methylated anthocyanin metabolites cross the blood-brain barrier via organic anion transporters.

• Accumulate in the CA1 and CA3 regions of the hippocampus and prefrontal cerebral cortex.

• Stimulate Extracellular Signal-Regulated Kinase (ERK1/2) and induce Brain-Derived Neurotrophic Factor (BDNF).

• Harvard Nurses' Health Study (n=16,010): High blueberry and strawberry intake delayed cognitive aging by up to 2.5 years.

Intestinal SGLT1/GLUT2 Transporter Inhibition

• Quercetin and anthocyanins competitively inhibit Sodium-Glucose Linked Transporter 1 (SGLT1) on the brush border membrane.

• Block glucose transporter 2 (GLUT2) basolateral translocation in enterocytes.

• Suppress postprandial blood glucose spikes by 25-33% following high-carbohydrate meals.

• Inhibit human salivary α-amylase, slowing complex starch breakdown kinetics in the upper duodenum.

Scientific Reference: Devore, E. E., Kang, J. H., Breteler, M. M., & Grodstein, F. (2012). Dietary intakes of berries and flavonoids in relation to cognitive decline. Annals of Neurology, 72(1), 135-143. DOI: 10.1002/ana.23594

Culinary & Medicinal Herbs: Rosmarinic Acid & Abietane Diterpenes

Key Biological Action: Lamiaceae species concentrate phenolic esters and volatile monoterpenes with extreme antioxidant capacity.

Lamiaceae Phenolic Terpenes & Essential Volatiles

• Rosmarinic Acid: Phenolic ester of caffeic acid with four active hydroxyl groups; suppresses complement C3b activation.

• Carnosic Acid & Carnosol (Rosemary): Highly lipophilic abietane diterpenes that break lipid peroxidation chains in cell membranes.

• Thymol & Carvacrol (Thyme/Oregano): Isomeric monoterpenes that permeabilize microbial cell walls, causing pathogen ATP leakage.

• Peppermint Menthol: Selective TRPM8 cold receptor agonist, inducing smooth muscle spasmolysis and local analgesia.

• Culinary Preservation: Acts as a natural lipid stabilizer in high-fat culinary and meat matrices.



Figure 5: Lamiaceae Botanical Matrix (Rosemary, Thyme, Oregano, Mint)

Scientific Reference: Paur, I., Balstad, T. R., & Blomhoff, R. (2011). Degree of antioxidant protection in herbs and spices. Herbal Medicine: Biomolecular and Clinical Aspects (2nd ed.), CRC Press, Ch. 4.

Curcumin & The Piperine Bioavailability Solution

Key Biological Action: Curcumin is a multi-target pleiotropic anti-inflammatory, unlocked 20-fold by black pepper piperine.

CURCUMIN ALONE (Low Bioavailability)

Serum AUC: < 1% Bioavailability

- Rapid intestinal & hepatic first-pass Phase II metabolism.
- Hepatic UDP-glucuronosyltransferase (UGT) converts >99% into inactive glucuronides.
- P-glycoprotein (ABCB1) pumps extrude free curcumin back into gut lumen.
- Clinical Result: Free serum curcumin remains undetectable (< 5 ng/mL).

CURCUMIN + 20mg PIPERINE (Synergy)

Serum AUC Surge: +2000% (20-Fold Increase)

- Piperine from black pepper non-competitively inhibits UGT & sulfation enzymes.
- Selectively suppresses P-glycoprotein efflux pumps in intestinal enterocytes.
- Safely pauses hepatic clearance of intact bioactive diferuloylmethane.
- Clinical Result: Delivers therapeutic free curcumin to target tissues (IKKβ, COX-2).

Human Pharmacokinetic Breakthrough: Human clinical trials (Shoba et al., 1998) established that 20 mg piperine co-administered with 2 g curcumin increases bioavailability by 2000% (20-fold), transforming curcumin from an excreted dietary compound into a potent systemic anti-inflammatory.

Scientific Reference: Shoba, G., Joy, D., Joseph, T., Majeed, M., Rajendran, R., & Srinivas, P. S. (1998). Influence of piperine on the pharmacokinetics of curcumin in animals and human volunteers. Planta Medica, 64(4), 353-356. DOI: 10.1055/s-2006-957450

Capsaicin & Gingerols: TRPV1 Activation & Anti-Emesis

Key Biological Action: Pungent vanilloid molecules activate thermo-sensitive ion channels and block neurokinin pathways.

Vanilloids, Gingerols & Neurosensory Targets

• Capsaicin (Chili Peppers): Selective agonist for Transient Receptor Potential Vanilloid 1 (TRPV1) calcium channels.

• Sensory Desensitization: Prolonged activation exhausts presynaptic Substance P stores, producing sustained clinical analgesia.

• Thermogenic Action: Promotes sympathetic catecholamine release, upregulating UCP-1 in brown adipose tissue (+8% metabolic rate).

• Gingerols & Shogaols (Ginger): Competitive antagonism of serotonin 5-HT3 and neurokinin NK1 receptors in the brainstem chemoreceptor trigger zone.

• Clinical Efficacy: Proven anti-emetic efficacy against pregnancy-induced nausea, motion sickness, and postoperative emesis.



Figure 6: High-Potency Spice Matrices (Turmeric, Ginger, Pepper, Chili)

Scientific Reference: Caterina, M. J., et al. (1997). The capsaicin receptor: a heat-activated ion channel in the pain pathway. Nature, 389(6653), 816-824. DOI: 10.1038/39807

Comprehensive Matrix: Bioactives, Receptors & Clinical Effect Sizes

Key Biological Action: Comparative synthesis of molecular mechanisms, metabolic fates, and verified clinical endpoints.

Plant Group	Key Bioactive	Primary Receptors	Cellular Action	Clinical Outcome
Cruciferous	Sulforaphane / DIM	Keap1 / AhR / HDACs	Nrf2-ARE induction, Phase II genes	45% reduction in genotoxic adducts
Tropical Fruits	Bromelain / Mangiferin	CD44 / COX-2 / AMPK	Proteolytic cleavage, GLUT4 uptake	Pain score -35%, glycemic blunting
Subtropical	Hesperidin / Punicalagin	eNOS / PINK1-Parkin	NO release, Urolithin mitophagy	Blood pressure -4.5 mmHg, muscle torque +12%
Temperate Fruits	Anthocyanins / Quercetin	ERK1/2 / SGLT1 / GLUT2	BDNF induction, carbohydrate block	Cognitive delay +2.5 yrs, postprandial glucose -28%
Herbs	Rosmarinic / Carnosic Acid	C3b complement / Nrf2	Lipid radical scavenging, anti-lysis	Reduced mucosal inflammation & oxidation
Spices	Curcumin / Capsaicin	IKKβ / NF-κB / TRPV1	Substance P depletion, thermogenesis	CRP reduction -40%, metabolic rate +8%

Scientific Reference: Williamson, G., & Manach, C. (2005). Bioavailability and bioefficacy of polyphenols in humans. American Journal of Clinical Nutrition, 81(1), 243S-255S. DOI: 10.1093/ajcn/81.1.243S

NUTRITIONAL OPTIMIZATION

Food Matrix Synergies, Bioavailability & Culinary Optimization

Key Biological Action: Co-ingesting lipophilic bioactives with dietary lipids and combining complementary spice hurdles maximizes uptake.

Turmeric + Pepper

Curcumin + Piperine

+2000% BIOAVAILABILITY

Piperine inhibits hepatic UGT glucuronidation, allowing free bioactive curcumin to reach systemic circulation.

Broccoli + Mustard

Glucoraphanin + Myrosinase

ENZYME RESCUE

Cooking broccoli denatures its myrosinase; adding raw ground mustard seeds supplies active myrosinase to restore sulforaphane release.

Green Tea + Citrus

EGCG + Ascorbic Acid

ACID STABILIZATION

Green tea catechins (EGCG) rapidly oxidize in alkaline intestines; citrus citric and ascorbic acids stabilize catechins, raising uptake by 4x.

Tomato + Olive Oil

Lycopene + Oleic Acid

+45% MICELLIZATION

Simmering tomatoes in extra virgin olive oil promotes trans-to-cis lycopene isomerization and mixed bile-salt micellar uptake.

Whole-Food Synergy Rule: Whole-food botanical pairings overcome digestive and hepatic absorption bottlenecks far more effectively than isolated single supplements. Co-ingesting lipid carriers with lipophilic carotenoids and spices boosts intestinal micellization by 300% to 500%.

Scientific Reference: Surh, Y. J. (2002). Anti-tumor promoting potential of selected spice ingredients. Food and Chemical Toxicology, 40(8), 1091-1097. DOI: 10.1016/S0278-6915(02)00078-9

Selected Peer-Reviewed Literature & Primary Citations

Key Biological Action: All biochemical pathways, clinical trials, and rate constants cited are fully traceable.

[1] Manach, C., Scalbert, A., et al. (2004). Polyphenols: food sources and bioavailability. Am. J. Clin. Nutr., 79(5), 727-747. DOI: 10.1093/ajcn/79.5.727

[2] Dinkova-Kostova, A. T., & Kostov, R. V. (2012). Glucosinolates and isothiocyanates in health and disease. Trends Mol. Med., 18(6), 337-347. DOI: 10.1016/j.molmed.2012.04.003

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[4] Kensler, T. W., et al. (2013). Keap1-nrf2 signaling: a target for cancer prevention. Carcinogenesis, 34(2), 243-252. DOI: 10.1093/carcin/bgs340

[5] Bradlow, H. L., et al. (1999). 2-hydroxyestrone: the 'good' estrogen. J. Cell. Biochem. Suppl., 32-33, 144-152.

[6] Pavan, R., Jain, S., Shraddha, & Kumar, A. (2012). Properties and therapeutic application of bromelain. Biotechnol. Res. Int., 2012, 976203. DOI: 10.1155/2012/976203

[7] Imran, M., et al. (2017). Mangiferin: a natural miracle bioactive compound. Lipids Health Dis., 16(1), 84. DOI: 10.1186/s12944-017-0449-y

[8] Gattuso, G., et al. (2007). Flavonoid composition of citrus juices. J. Agric. Food Chem., 55(25), 9989-1002. DOI: 10.1021/jf071700a

[9] Espín, J. C., et al. (2013). Biological significance of urolithins. Mol. Nutr. Food Res., 57(7), 1199-1213. DOI: 10.1002/mnfr.201300058

[10] Devore, E. E., et al. (2012). Dietary intakes of berries in relation to cognitive decline. Ann. Neurol., 72(1), 135-143. DOI: 10.1002/ana.23594

[11] Shoba, G., et al. (1998). Influence of piperine on the pharmacokinetics of curcumin. Planta Med., 64(4), 353-356. DOI: 10.1055/s-2006-957450

[12] Caterina, M. J., et al. (1997). The capsaicin receptor: a heat-activated ion channel. Nature, 389(6653), 816-824. DOI: 10.1038/39807